

NORTH SYDNEY GIRLS HIGH SCHOOL

Section I: Multiple Choices

5 marks

Attempt all questions.

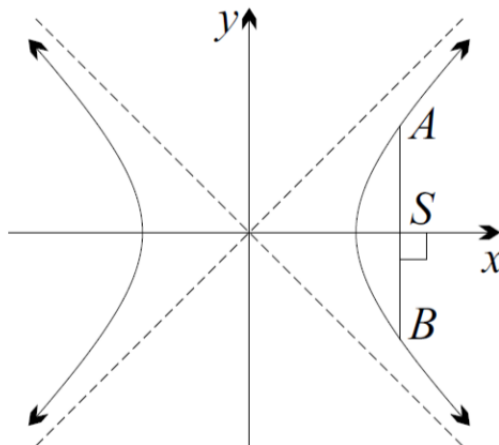
Answer on the multiple choice answer sheet provided for Section I.

1. What type of curve does the equation $|z - 3| - |z + 3| = 4$ corresponds to in the Argand diagram?
- (A) Circle
(B) Ellipse
(C) Parabola
(D) Hyperbola
2. The point $T(a \cos \theta, a \sin \theta)$ lies on the circle $x^2 + y^2 = a^2$. Which of the following is the equation of the tangent at T ?
- (A) $x \cos \theta + y \sin \theta = a$
(B) $x \cos \theta + y \sin \theta = a^2$
(C) $x \cos \theta - y \sin \theta = a$
(D) $x \cos \theta - y \sin \theta = a^2$
3. Given that $\sin(A - B)x + \sin(A + B)x = 2 \sin Ax \cos Bx$, which of the following is equal to

$$\int \sin 3x \cos 2x \, dx ?$$

- (A) $\frac{1}{2} \sin 5x + \frac{1}{2} \sin x + C$
(B) $\frac{1}{10} \cos 5x + \frac{1}{2} \cos x + C$
(C) $-\frac{1}{2} \sin 5x - \frac{1}{2} \sin x + C$
(D) $-\frac{1}{10} \cos 5x - \frac{1}{2} \cos x + C$

4. Which of the following are the foci of the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$?
- (A) $(2,0)$ and $(-2,0)$
- (B) $(3\sqrt{5},0)$ and $(-3\sqrt{5},0)$
- (D) $\left(\frac{3\sqrt{5}}{2},0\right)$ and $\left(-\frac{3\sqrt{5}}{2},0\right)$
- (C) $(\sqrt{5},0)$ and $(-\sqrt{5},0)$
5. The graph of the hyperbola $\frac{x^2}{9} - \frac{y^2}{5} = 1$ is shown below. S is one of the foci and AB is a chord of the hyperbola that is perpendicular to the x -axis at S .



What is the length of AB ?

- (A) $\frac{10}{3}$
- (B) $\frac{18}{\sqrt{5}}$
- (C) $\frac{10}{9}$
- (D) $\frac{18}{5}$

END OF SECTION I

Section II

Attempt all questions.

Answer each question in a separate writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (16 marks)

(a) Evaluate $\int_0^7 \frac{2}{\sqrt{x+9}} dx$. 2

(b) Find $\int \sin^3 x \, dx$ 3

(c) (i) Write $\frac{2x+3}{(x-2)(x^2+2)}$ in the form $\frac{A}{x-2} + \frac{Bx+C}{x^2+2}$ 2

(ii) Hence find $\int \frac{2x+3}{(x-2)(x^2+2)} dx$ 3

(d) Find $\int \frac{dx}{\sqrt{13+4x-x^2}}$ 3

(e) Find $\int x \cos 2x \, dx$ 3

End of Question 6

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Question 7 (20 marks)

- (a) The point $P(4\sec\theta, 3\tan\theta)$ lies on a hyperbola that has both vertices on the x -axis.
- (i) Find the equation of the hyperbola in Cartesian form. 1
- (ii) Find the eccentricity of the hyperbola. 1
- (iii) Draw a neat **HALF** page sketch of this hyperbola, showing the coordinates of the vertices and the foci, and the equations of the directrices and asymptotes. 3
- (iii) Derive the equation of the tangent to the hyperbola at the point P in parametric form. 2

- (b) The points $P\left(2p, \frac{2}{p}\right)$ and $Q\left(2q, \frac{2}{q}\right)$ lie on the rectangular hyperbola $xy = 4$.

M is the midpoint of PQ .

P and Q move on the hyperbola so that the chord PQ will always pass through the point $R(4,2)$ and $pq > 0$

- (i) Show that the chord PQ has the equation $x + pqy = 2(p + q)$. 2
- (ii) Show that $pq = p + q - 2$. 1
- (iii) Hence find the equation of the locus of M as P moves on the hyperbola. 3

Question 7 (continued)

(c) (i) Using the substitution $x = \tan \frac{1}{2} x$ or otherwise, show that $\int_0^{\frac{\pi}{2}} \frac{1}{1 - \frac{1}{2} \sin x} dx = \frac{4\pi}{3\sqrt{3}}$ **3**

(ii) Show that $\int_0^{2a} f(x) dx = \int_0^a [f(x) + f(2a - x)] dx$ **2**

(iii) Hence evaluate $\int_0^{\pi} \frac{x}{1 - \frac{1}{2} \sin x} dx$ **2**

END OF EXAMINATION

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SECTION I**Q1 D****Q2 A****Q3 D****Q4 C****Q5 A****SECTION II****Question 16****(a)**

$$\begin{aligned}\int_0^7 \frac{2}{\sqrt{x+9}} dx &= \int_0^7 2(x+9)^{-\frac{1}{2}} dx \\ &= \left[4(x+9)^{\frac{1}{2}} \right]_0^7 = \left[4\sqrt{x+9} \right]_0^7 \\ &= \left[(4\sqrt{7+9}) - (4\sqrt{0+9}) \right] \\ &= 4\end{aligned}$$

(b)

$$\begin{aligned}\int \sin^3 x dx &= \int \sin x \sin^2 x dx \\ &= \int \sin x (1 - \cos^2 x) dx \\ &= \int \sin x - \sin x \cos^2 x dx \\ &= -\cos x + \frac{\cos^3 x}{3} + C\end{aligned}$$

(c) (i)

$$\begin{aligned}\frac{2x+3}{(x-2)(x^2+2)} &= \frac{A}{x-2} + \frac{Bx+C}{x^2+2} \\ 2x+3 &= A(x^2+2) + (Bx+C)(x-2)\end{aligned}$$

Sub $x = 2$

$$7 = A(4+2)$$

$$A = \frac{7}{6}$$

Sub $x = 0$

$$3 = A(0+2) + C(-2)$$

$$3 = \frac{7}{6}(2) + C(-2)$$

$$C = -\frac{1}{3}$$

Sub $x = 1$

$$5 = A(1+2) + (B+C)(-1)$$

$$5 = 3\left(\frac{7}{6}\right) - B - \frac{1}{3}(-1)$$

$$B = \frac{7}{3} - 5 + \frac{1}{3}$$

$$B = -\frac{7}{6}$$

$$\frac{2x+3}{(x-2)(x^2+2)} = \frac{\frac{7}{6}}{x-2} + \frac{-\frac{7}{6}x - \frac{1}{3}}{x^2+2}$$

(ii) ho

$$\begin{aligned}\int \frac{2x+3}{(x-2)(x^2+2)} dx &= \int \frac{\frac{7}{6}}{x-2} dx + \int \frac{-\frac{7}{6}x - \frac{1}{3}}{x^2+2} dx \\ &= \frac{7}{6} \int \frac{1}{x-2} dx + \int \frac{-\frac{7}{6}x}{x^2+2} + \int \frac{-\frac{1}{3}}{x^2+2}\end{aligned}$$

$$\begin{aligned}&= \frac{7}{6} \int \frac{1}{x-2} dx - \frac{7}{12} \int \frac{2x}{x^2+2} dx - \frac{1}{3} \int \frac{1}{x^2+2} dx \\ &= \frac{7}{6} \ln(x-2) - \frac{7}{12} \ln(x^2+2) + \frac{1}{3\sqrt{2}} \tan^{-1} \frac{x}{\sqrt{2}} + C\end{aligned}$$

(d)

$$\begin{aligned}\int \frac{dx}{\sqrt{13+4x-x^2}} &= \int \frac{dx}{\sqrt{17-(x^2-4x+4)}} \\ &= \int \frac{dx}{\sqrt{17-(x-2)^2}} \\ &= \sin^{-1} \frac{x-2}{\sqrt{17}} + C\end{aligned}$$

(e)

$$\begin{aligned}\int x \cos 2x dx &= \frac{x \sin 2x}{2} - \int \frac{\sin 2x}{2} dx \\ &= \frac{x \sin 2x}{2} + \frac{\cos 2x}{4}\end{aligned}$$

Question 7**a)****(i)** Find the equation of the hyperbola in Cartesian form.

$$(4 \sec \theta, 3 \tan \theta)$$

$$a = 4 \text{ and } b = 3$$

$$\text{Equation is } \frac{x^2}{16} - \frac{y^2}{9} = 1$$

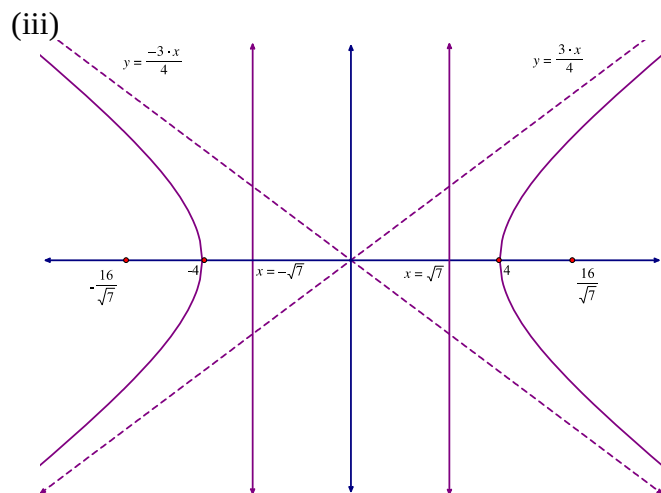
(ii) Find the eccentricity of the hyperbola.

$$b^2 = a^2(1 - e^2)$$

$$\frac{9}{16} = 1 - e^2$$

$$\frac{7}{16} = e^2$$

$$\frac{\sqrt{7}}{4} = e$$



(iv) Derive the equation of the tangent to the hyperbola at the point P in parametric form.

At $P(4\sec\theta, 3\tan\theta)$

By Implicit Differentiation

$$\frac{2x}{16} - \frac{2y}{9} \times \frac{dy}{dx} = 0$$

$$\frac{2y}{9} \times \frac{dy}{dx} = \frac{2x}{16}$$

$$\frac{dy}{dx} = \frac{9x}{16y}$$

At $P(4\sec\theta, 3\tan\theta)$

$$\frac{dy}{dx} = \frac{3\sec\theta}{4\tan\theta}$$

Equation of the tangent at P

At $P(4\sec\theta, 3\tan\theta)$

$$y - 3\tan\theta = \frac{3\sec\theta}{4\tan\theta}(x - 4\sec\theta)$$

$$4y\tan\theta - 12\tan^2\theta = 3x\sec\theta - 12\sec^2\theta$$

$$4y\tan\theta = 3x\sec\theta - 12\sec^2\theta + 12\tan^2\theta$$

$$4y\tan\theta = 3x\sec\theta + 12$$

b)

(i) Show that the chord PQ has the equation

$$x + pqy = 2(p + q)$$

Gradient of PQ

$$m_{PQ} = \frac{\frac{2}{p} - \frac{2}{q}}{2p - 2q}$$

$$= \frac{2(q - p)}{2(p - q)}$$

$$= -\frac{1}{pq}$$

Equation of PQ

$$y - \frac{2}{p} = -\frac{1}{pq}(x - 2p)$$

$$pqy - 2q = -x + 2p$$

$$x + pqy = 2(p + q)$$

(ii) Show that $pq = p + q - 2$.

At $R(4, 2)$ Let $x = 4$ and $y = 2$

$$x + pqy = 2(p + q)$$

$$(4) + pq(2) = 2(p + q)$$

$$2 + pq = (p + q)$$

$$pq = (p + q) - 2$$

(iii) Hence find the equation of the locus of M as P moves on the hyperbola.

Midpoint of PQ

$$x = \frac{2p + 2q}{2}$$

$$x = p + q$$

$$y = \frac{\frac{2}{p} + \frac{2}{q}}{2}$$

$$y = \frac{p + q}{pq}$$

Sub $x = p + q$ and $pq = (p + q) - 2$ into

$$y = \frac{p+q}{pq}$$

$$y = \frac{x}{(p+q)-2}$$

$$= \frac{x}{x-2}$$

But $pq > 0$ so P and Q must be on the same branch and as PQ must pass through R then both p and q must be positive.

Thus the equation of the locus of M is

$$y = \frac{x}{x-2} \quad x > 2$$

Question 7 (continued)

(c) (i) Using the substitution $x = \tan \frac{x}{2}$ or

otherwise, show that $\int_0^{\frac{\pi}{2}} \frac{1}{1 - \frac{1}{2} \sin x} dx = \frac{4\pi}{3\sqrt{3}}$

$$\int_0^{\frac{\pi}{2}} \frac{1}{1 - \frac{1}{2} \sin x} dx = \int_0^1 \frac{1}{1 - \frac{1}{2} \times \frac{2t}{1+t^2}} \frac{2dt}{1+t^2} dx$$

$$= \int_0^1 \frac{2dt}{1+t^2-t}$$

$$= \int_0^1 \frac{2dt}{1 - \frac{1}{4} + t^2 - t + \frac{1}{4}}$$

$$= 2 \int_0^1 \frac{dt}{\frac{3}{4} + \left(t - \frac{1}{2}\right)^2}$$

$$= 2 \left[\frac{2}{\sqrt{3}} \tan^{-1} \frac{2\left(t - \frac{1}{2}\right)}{\sqrt{3}} \right]_0^1$$

$$= \frac{4}{\sqrt{3}} \left[\tan^{-1} \frac{2\left(\frac{1}{2}\right)}{\sqrt{3}} - \tan^{-1} \frac{2\left(-\frac{1}{2}\right)}{\sqrt{3}} \right]$$

$$= \frac{4}{\sqrt{3}} \left[\frac{\pi}{6} - \left(-\frac{\pi}{6}\right) \right]$$

$$= \frac{4\pi}{3\sqrt{3}} \quad \quad \quad \mathbf{3}$$

(ii) Show that

$$\int_0^{2a} f(x) dx = \int_0^a [f(x) + f(2a-x)] dx$$

$$LHS = \int_0^{2a} f(x) dx$$

$$= \int_0^a f(x) dx + \int_a^{2a} f(x) dx$$

and

$$\int_a^{2a} f(x) dx = \int_0^a f(2a-x) dx \text{ by reflecting in the line } x = a$$

OR

$$\text{Let } u = 2a - x$$

$$du = -dx$$

$$\text{When } x = a \quad u = a$$

$$\text{When } x = 2a \quad u = 0$$

$$\int_a^{2a} f(u) du = \int_0^a f(2a-x) dx$$

$$LHS = \int_0^a f(x) dx + \int_a^{2a} f(x) dx$$

$$= RHS$$

$$\begin{aligned}
\int_0^{\pi} \frac{x}{1-\frac{1}{2}\sin x} dx &= \int_0^{\frac{\pi}{2}} \frac{x}{1-\frac{1}{2}\sin x} + \frac{\pi-x}{1-\frac{1}{2}\sin(\pi-x)} dx \\
&= \int_0^{\frac{\pi}{2}} \frac{x}{1-\frac{1}{2}\sin x} dx + \int_0^{\frac{\pi}{2}} \frac{\pi}{1-\frac{1}{2}\sin(\pi-x)} dx - \int_0^{\frac{\pi}{2}} \frac{x}{1-\frac{1}{2}\sin(\pi-x)} dx \\
&= \int_0^{\frac{\pi}{2}} \frac{x}{1-\frac{1}{2}\sin x} dx + \int_0^{\frac{\pi}{2}} \frac{\pi}{1-\frac{1}{2}\sin x} - \int_0^{\frac{\pi}{2}} \frac{x}{1-\frac{1}{2}\sin x} dx \\
&= \int_0^{\frac{\pi}{2}} \frac{\pi}{1-\frac{1}{2}\sin x} dx \\
&= \pi \int_0^{\frac{\pi}{2}} \frac{1}{1-\frac{1}{2}\sin x} dx = \frac{4\pi^2}{3\sqrt{3}}
\end{aligned}$$